

Final Project

SPSS For Research

Instructor: Md. Ahsanul Islam

Deadline: 19 Sep 2026

The final project is your opportunity to use a real dataset. This is an individual project.

Reproducibility is required in this project, so make sure to save your Syntax file.

1. Dataset selection

Select ONE dataset from the approved sources below. You must have your dataset and research question approved by the instructor before the proposal-approval date; unapproved topics will not be accepted.

Source	What it offers / why it is suitable	Link
Bangladesh DHS (DHS Program)	Nationally representative health and demographic survey with hundreds of variables (maternal & child health, nutrition, wealth, education).	https://dhsprogram.com/data/dataset/Bangladesh_Standard-DHS_2022.cfm
UNICEF MICS	Multiple Indicator Cluster Surveys. Contains data on child health, water & sanitation, education, women's status. It is well documented, ideal for group-comparison and logistic-regression questions.	https://mics.unicef.org/surveys?display=card&f[0]=round:52&f[1]=region:3781&f[2]=status:241
UCI Machine Learning Repository	Curated teaching datasets across health, social science and business, each with a documented codebook.	https://archive.ics.uci.edu/
Kaggle Datasets	Large collection of open datasets. Permitted only if the dataset has a clear source, a codebook, and meets the size and variable criteria below.	https://www.kaggle.com/datasets

Your dataset must meet all of the following criteria:

- At least 6 usable variables, with a mix of categorical and continuous types.
- At least 150 cases (rows).
- Contains one binary/categorical outcome suitable for logistic regression, and at least one continuous variable suitable for a t-test/ANOVA and for correlation or linear regression.

- Publicly available and properly citable. Should not contain any personally identifying information such as phone number, name, etc..
- Some realistic data issues to clean (missing values, inconsistent coding, or duplicates) will help demonstrate your data cleaning skill.

2. What you must do

- (1) State one primary research question and one or two secondary questions. For each, write the null and alternative hypotheses in words.
- (2) Properly import the data and define variable properties (labels, value labels, measurement level). Declare missing-value codes, recode/reverse-code where needed, remove duplicate records, and handle anomalies. Report how many cases remain.
- (3) Summaries the sample in a “Table 1”, frequency and percentage for categorical variables, and mean (SD), median, minimum and maximum for continuous variables.
- (4) Produce at least three appropriate, well-labelled graphs (for example a histogram, a bar chart, a boxplot and/or a scatterplot) that identify the characteristics of your selected variables.
- (5) Answer at least one research question with an appropriate group-comparison or association test (independent-samples t-test, one-way ANOVA with post-hoc, paired t-test, chi-square or non-parametric test). Check the relevant assumptions (normality, homogeneity of variance) and report the result in APA format.
- (6) Fit a regression model appropriate to your outcome: a binary logistic regression for a binary outcome, or a multiple linear regression for a continuous outcome. Justify the predictors, handle categorical predictors correctly, examine diagnostics or model-fit statistics, and interpret the coefficients (or odds ratios) in plain language.
- (7) Prepare everything into a clear written report with correct APA-style tables and a conclusion that answers your research question(s) in a few sentences.

3. Report structure

Your written report should follow this structure:

- **Title and name** (as per certificate).
- **Introduction / background:** Introduce the topic, the dataset, and your research question(s) with hypotheses. (No need to perform literature review)
- **Methods:** Dataset source, sample, variables and the cleaning and analysis steps you performed.
- **Results:** Table 1, graphs, table of the hypothesis test(s), and table of the fitted model. Each should be reported in APA style.
- **Discussion / conclusion:** What the results mean, limitations, and the answer to your research question(s).
- **References:** The dataset citation and any sources, in APA style.

4. Submission requirements

Submit a single zip file named **finalproject_YOURNAME.zip** to Google Classroom, containing:

- The written report as **finalproject_report_YOURNAME.docx** (or PDF).
- The cleaned SPSS dataset, **finalproject_data_YOURNAME.sav**.
- The syntax file, **finalproject_syntax_YOURNAME.sps**.

5. Marking rubric (100 marks)

Component	What earns the marks	Marks
Research question selection	Clear research question(s) with correctly stated null and alternative hypotheses.	5
Dataset selection	Appropriate, well-justified dataset	5
Data management & cleaning	Codebook and variable properties defined, missing values, recoding, reverse-coding and duplicates handled correctly, every step reproducible from syntax.	15
Descriptive statistics	Correct summary of the sample in a clear Table 1 (frequency/%, mean (SD), median, min, max as appropriate).	10
Data visualization	Appropriate, publication-ready graphs (e.g. histogram, bar chart, boxplot, scatterplot) that support the analysis, correctly labelled.	15
Hypothesis testing	Correct choice of test (t-test / ANOVA / chi-square / correlation), assumptions checked, results reported in APA format.	15
Regression Modeling	Correctly specified model, categorical predictors handled correctly, diagnostics or model-fit reported, regression / odds ratios interpreted correctly.	20
Reporting & writing quality	Clear structure, APA-style tables and text, accurate interpretation and a sensible conclusion.	10
Reproducibility & submission	Syntax runs end-to-end; dataset, syntax and report all included and named as required.	5
Total		100